What does renal failure teach us about our National Health System?

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After 45 years, I am retiring from the Italian National Health Service. If someone were to ask me what I think of the Italian National Health Service after all these years, I would consider Brenner's theory on single nephron hyperfiltration injury.[1] In chronic kidney disease, the progressive loss of functional nephrons—caused by conditions like hypertension and diabetes—forces the remaining nephrons to compensate through hyperfiltration and increased workload. This adaptive mechanism, mainly mediated by angiotensin II, temporarily sustains kidney function but ultimately accelerates damage. Eventually, as the surviving nephrons become overburdened and exhausted, the inevitable decline towards end-stage kidney disease occurs.

In my view, this pathophysiological sequence parallels the state of the Italian National Health Service and beyond. Just as a nephron struggles under increasing pressure, an overstretched health-care system (burdened by growing demands and systemic inefficiencies) faces progressive deterioration. Without intervention, both ultimately collapse under the weight of chronic overload.

For the **Italian National Institute of Statistics** see https://esploradati.istat.it

The Italian National Health Service faces excessive bureaucratic burdens, inefficient coordination between hospitals and territorial services, and poor integration among social services, general practitioners, and local specialists. The core functional units of the Italian National Health Service—hospital doctors, nurses, and general practitioners—are struggling with the burden of an ageing population with comorbidities. Hospitals are overwhelmed, placing an unsustainable strain on health-care professionals, who, frustrated and underpaid, are leaving the National Health Service in increasing numbers. Those who remain, similar to overworked nephrons, are left to manage relentless clinical duties. General practitioners face a similar crisis, strained by excessive bureaucracy. Akin to the progressive loss of glomeruli in a failing kidney, the system is critically understaffed, with a shortage of approximately 30 000 hospital doctors and 70 000 nurses.[3]

For patients with end-stage kidney disease, dialysis or transplantation can be lifesaving. Unfortunately, no equivalent interventions exist for a National Health System, unless profound reforms take place. The available options are either accepting long waiting lists and financial constraints or, for the wealthiest, turning to the private sector, thus worsening health-care inequality.

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A new threat to obesity: vanity sizing

The global burden of obesity is expanding worldwide.[1] Obesity is influenced by cultural and social factors, with wide variation across countries. These factors include food preferences, societal standards for body shape, and practices around the use of leisure time and physical activity, all of which can interact with other risk factors to drive increasing obesity rates. In this Correspondence, we call attention to how the apparel industry can negatively influence the perception of obesity in the general population by downplaying its effect on health.

Clothing is a necessity, making the apparel industry essential to the world economy. A decade ago, controversy in the fashion world was focused on the lack of sizes for people with overweight and obesity, which contributed to weight-related disease and reinforced social inequality. However, due to social pressure and social media (ie, new influencers for society) this controversy has been somewhat resolved and fortunately; nowadays, many brands are becoming more size inclusive, although this can certainly be improved.[2]

The teenage daughter of one of the authors explains that: "Last year I wore a 36 and now I wear a 32 because the 34 is huge...and I have grown 5 cm (1·96 in)." She, of course, refers to clothing sizing, which is how it is measured in Spain. In other instance, a 45-year-old friend of an author with overweight expressed confusion about how now she fits into a smaller size of the same clothing brand, despite weighing more than 6 months earlier.

Let us examine this second scenario. In recent years, the apparel industry has implemented a series of measures to satisfy consumers by making them believe that despite gaining weight, their size remains the same or has even decreased. This concept is known as vanity sizing and involves labelling a clothing item with a size smaller than its actual size.[3] This practice is common among mainstream clothing chains and is a way to promote brand loyalty and drive consumer spending via misleading tactics that generate confusion.[4] In such instances, the promotion of social stereotypes comes into play, which contributes to psychological distress. A major obstacle to eliminating vanity sizing is the absence of an international sizing standard. Do you always wear the same size no matter which store you shop at? Most likely not. Clothing sizes are inconsistent and confusing. Ideally, fashion consortiums should agree to adopt a universal sizing to simplify shopping and avoid discrepancies, but the reality is that each brand sets its own measurements. Health-care professionals should actively raise awareness of this issue and urge industry leaders and policy makers to prioritise consumer wellbeing over marketing strategies.

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The need for a type 1 diabetes scorecard

The predicted global rise in diabetes prevalence, coupled with related mortality and inequity, has led WHO to launch the Global Diabetes Compact.[1] This initiative sets evidence-based targets for global diabetes care, emphasising universal access to affordable insulin and glucose monitoring for individuals with type 1 diabetes, and for achieving adequate glycaemic control in at least 80% of people with diabetes by 2030. Achieving these targets requires public health strategies for expanding access to health-care systems, insulin, disease management tools, and education to address the challenges of living with type 1 diabetes. This work also requires robust indicators to monitor progress, measure outcomes, and evaluate cost-effectiveness. In fact, the notion of an indicator (ie, a quantitative metric used to assess performance, measure achievement, and ensure accountability) is essential. Unfortunately, diabetes outcome indicators at an international (and national) level are still limited and provide insufficient information on the cost-effectiveness of interventions specific to diabetes.[2]

An alternative source of information can be generated by people with type 1 diabetes who routinely collect data on disease management to guide provider feedback and diabetes management. Metrics such as insulin access, self-monitoring data (eg, continuous glucose monitoring), educational needs, and glycaemic control (eg, HbA_{1c} concentrations and time in range) offer insights into the effectiveness of self-management. Documenting complications and their risk factors also facilitates both early and secondary prevention. Beyond metabolic indicators, patient-reported outcomes and quality-of-life metrics can also be collected and will highlight the bidirectional relationship between wellbeing and clinical outcomes.

A type 1 diabetes scorecard, aggregating multiple indicators from different sources, could provide a comprehensive appraisal of the disease landscape at specific timepoints and locations. Longitudinal data collection would track changes over time, offering insights for both immediate and future analyses. To be effective, the scorecard should be affordable, user-friendly, and insightful for all stakeholders, facilitating global monitoring and management of type 1 diabetes. Sustainable governance and funding are needed to ensure that the indicators remain fit for purpose.[3]

Advances in electronic health records could streamline data collection, reducing time and cost burdens, and enable standardised data collection for benchmarking comparisons across treatment centres, informing education and treatment strategies.[4] Further harmonisation of real-word data collection will increase our understanding of what strategies actually work[5] to truly assess global progress in the management of type 1 diabetes. Diabetes care structures vary widely from tertiary care teams with advanced technologies to community workers providing essential survival kits in remote regions. In resource-limited settings, high workloads might constrain data collection to minimal datasets. Solutions must account for these disparities, ensuring the inclusion of outcome data from all care contexts. Although tools such as the T1D Index provide valuable country-level data from peer-reviewed publications, they are limited by infrequent updates.[6] We expect that use of a scorecard would provide a regular update for the T1D Index.

For more on the **T1D Index** see https://www.t1dindex.org/

Additionally, for sustainability, diabetes interventions need support from policy makers and integration into policy planning and budgeting, which will require additional indicators, including cost and cost-effectiveness data. Although population-based surveys and administrative data (eg, hospitalisation rates, supply chain metrics, reimbursement data, and mortality rates) offer valuable insights for public health assessments, these indicators do not always identify the specific causes behind the observed changes.

In sum, a type 1 diabetes scorecard would enable countries to monitor health-care systems and diabetes